$$\mathcal{D} = \{(x_i, y_i)\}_{i=1}^{N}$$

$y = f(x)$

$$B = \{f_1, f_2, \cdots, f_k\}$$

$$f \simeq \alpha_1 f_1 + \alpha_2 f_2 + \cdots + \alpha_k f_k \quad : \text{model}$$

$$\alpha = \begin{pmatrix} \alpha_1 \\ \vdots \\ \alpha_k \end{pmatrix} \leftarrow \text{parameters / weights}$$

Loss fn: $r^T r = \sum_{i=1}^{N} r_i^2 = \sum_{i=1}^{N} (y_i - \hat{y}_i)^2$

$$= \sum_{i=1}^{N} (y_i - \alpha_1 f_1(x_i) - \alpha_2 f_2(x_i) - \cdots - \alpha_k f_k(x_i))^2$$

$$= (y - \Phi\alpha)^T (y - \Phi\alpha)$$

$$\min_{\alpha \in \mathbb{R}^k} (y - \Phi\alpha)^T (y - \Phi\alpha)$$

$\|\cdot\|_2 \Leftarrow$ length of a vector.

$$= \min_{\alpha \in \mathbb{R}^k} \|y - \Phi\alpha\|_2^2$$

Observation:

There may exist an $\hat{\alpha} \in \mathbb{R}^k$ s.t. $\|y - \Phi\hat{\alpha}\|_2^2 = 0$

$\Leftrightarrow \exists\ \hat{\alpha}$ s.t. $\Phi\hat{\alpha} = y$

The system of eqns $\Phi\alpha = y$ has a solution.

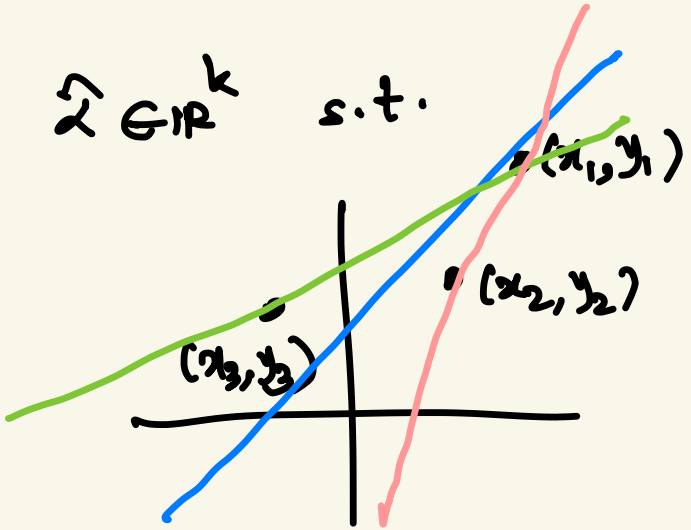
Given $\mathcal{D}$, construct $\Phi, \alpha, y, \hat{y}$.

If $\Phi\alpha = y$ has a solution, then the minimum value of loss $\neq$ zero.

Case(ii) The minimum value of L is **NOT** zero.

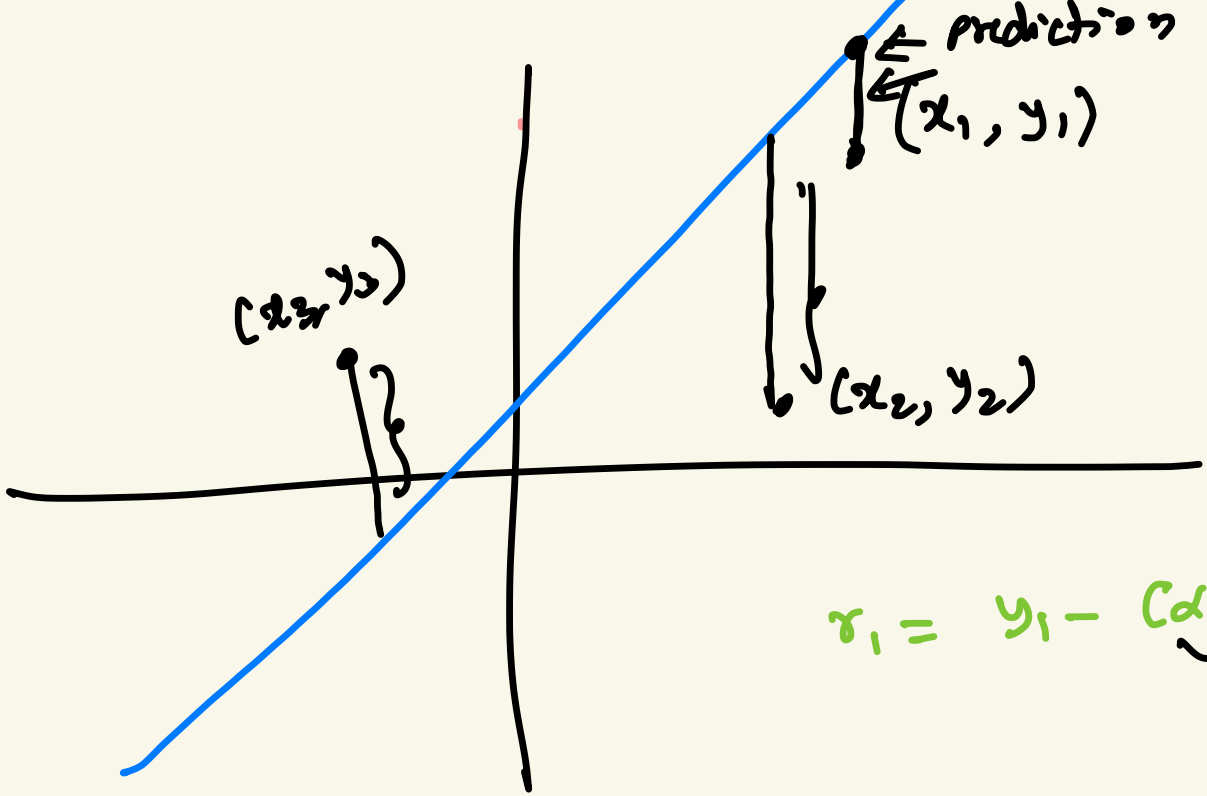
There does not exist any $\hat{\alpha} \in \mathbb{R}^k$ s.t.

$$\| y - \Phi\hat{\alpha} \|_2^2 = 0$$



Basis functions $\{1, x\}$.

$$f \approx \alpha_1 + \alpha_2 x$$



$$r_1 = y_1 - \underbrace{(\alpha_1 + \alpha_2 x_1)}_{\hat{y}_1}$$

There does NOT exist a solution to the system of eqⁿ:

$$\underline{\Phi} \in \mathbb{R}^{\underline{N} \times \underline{k}}$$

$$\underline{\Phi} \alpha = y$$

$$\Rightarrow y \notin \text{span}\{\underbrace{\phi_1, \phi_2, \ldots, \phi_k}_{\text{columns of } \underline{\Phi}}\}$$

$$y \notin R(\underline{\Phi}) \quad (\text{range space of } \underline{\Phi})$$

$\alpha \in \mathbb{R}^2$

$\alpha_1 \phi_1 + \alpha_2 \phi_2$

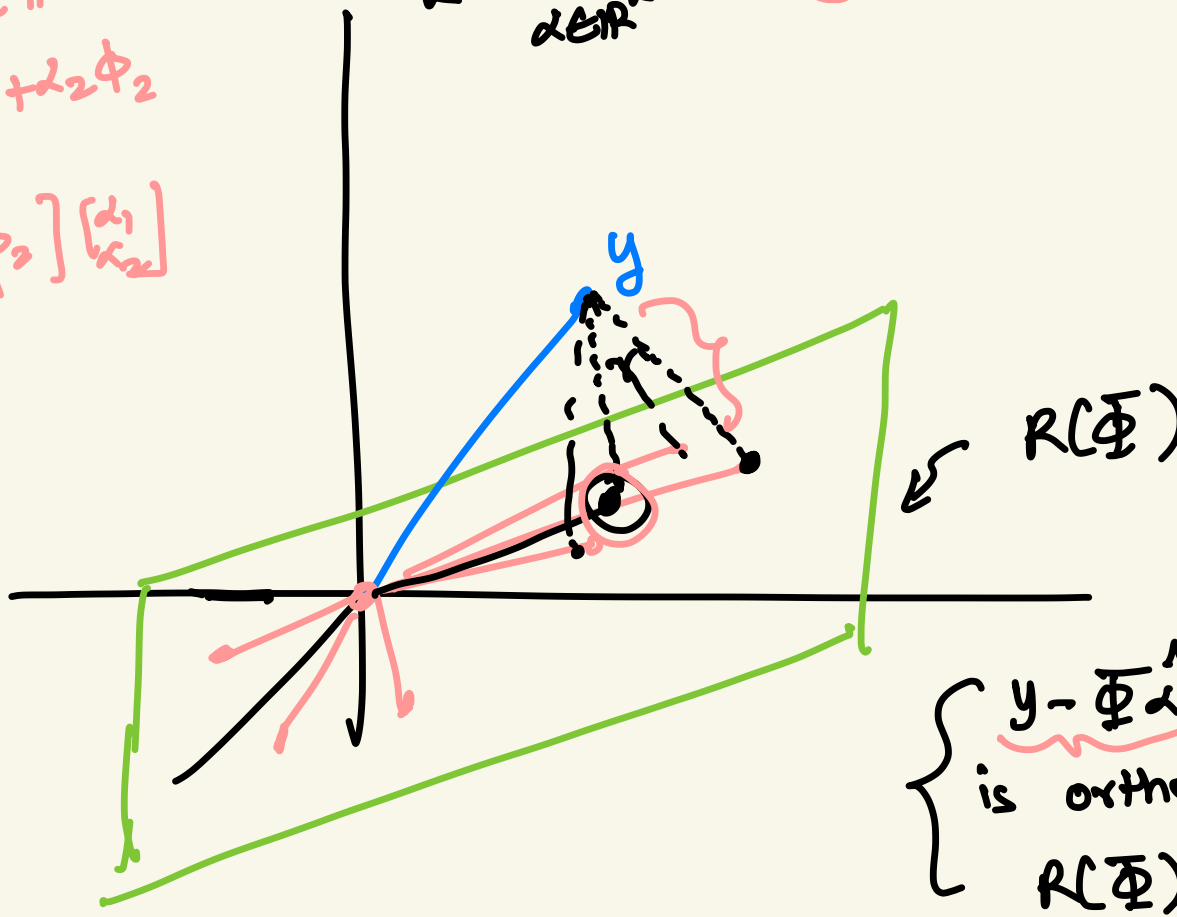
$\begin{bmatrix} | & | \\ \phi_1 & \phi_2 \\ | & | \end{bmatrix} \begin{bmatrix} \alpha_1 \\ \alpha_2 \end{bmatrix}$

$\mathcal{L} = \min_{\alpha \in \mathbb{R}^k} \| y - \Phi \alpha \|_2^2$

$\mathbb{R}^3$

$\Phi \in \mathbb{R}^{3 \times 2}$

$\Phi = [\phi_1, \phi_2]$



$\left\{ \begin{array}{l} y - \Phi \hat{\alpha} \in \text{residual} \\ \text{is orthogonal to} \\ R(\Phi) \end{array} \right\}$

Note that choose $\hat{\alpha} \in \mathbb{R}^k$ s.t. the residual $y - \Phi\hat{\alpha}$ is orthogonal to $R(\Phi)$

$$y - \Phi\hat{\alpha} \perp \Phi z \quad \forall z \in \mathbb{R}^k$$

$$\Rightarrow \langle y - \Phi\hat{\alpha}, \Phi z \rangle = 0 \quad \forall z \in \mathbb{R}^k$$

$$\Rightarrow (\Phi z)^T (y - \Phi\hat{\alpha}) = 0 \quad \forall z \in \mathbb{R}^k$$

$$\Rightarrow z^T \Phi^T (y - \Phi\hat{\alpha}) = 0$$

$$\Rightarrow z^T \left[\Phi^T (y - \Phi\hat{\alpha})\right] = 0 \quad \forall z \in \mathbb{R}^k$$

$$\Phi e_1 = \phi_1, \quad \Phi e_i = \phi_i$$

$$\{ z_1\phi_1 + z_2\phi_2 + \cdots + z_k\phi_k : z_i \in \mathbb{R} \}$$

$$\| \quad \Phi z$$

$$\begin{bmatrix} \phi_1 & \phi_2 & \cdots & \phi_k \end{bmatrix} \begin{pmatrix} z_1 \\ z_2 \\ \vdots \\ z_k \end{pmatrix}$$

$$= z_1\phi_1 + \cdots + z_k\phi_k$$

Inner product of $x, y \in \mathbb{R}^k$

$$\langle x, y \rangle = y^T x = x^T y = x_1 y_1 + \cdots + x_k y_k$$

$$x = \begin{bmatrix} x_1 \\ \vdots \\ x_k \end{bmatrix}, \quad y = \begin{bmatrix} y_1 \\ \vdots \\ y_k \end{bmatrix}$$

Two vector x & y are othogonal,

$$\langle x, y \rangle = 0$$

For every $z \in \mathbb{R}^k$

$$\langle z, x \rangle = 0$$

$$z^T x = 0$$

$$\Rightarrow \underline{x = 0} \quad \text{why?!}$$

Take $z = x \Rightarrow x^T x = 0$

$$\Rightarrow \sum x_i^2 = 0$$

$$\Rightarrow x_i = 0 \ \forall i.$$

$$\Rightarrow x = 0$$

$$\Rightarrow \Phi^T (y - \Phi \hat{\alpha}) = 0$$

$$\Rightarrow \Phi^T \Phi \hat{\alpha} = \Phi^T y$$

$\Rightarrow \hat{\alpha}$ is a solution to the system of eqns

$$\Phi^T \Phi \alpha = \Phi^T y$$

$\mathcal{D} = \{(x_i, y_i)\}_{i=1}^N$ $\quad$ $y = f(x)$ $\quad$ $f_1, f_2, \dots, f_k : \mathbb{R}^{(d)} \to \mathbb{R}$, $x_i \in \mathbb{R}^n$

$\mathcal{B} = \{f_1, f_2, \dots, f_k\}$ $\quad$ $f \approx \alpha_1 f_1 + \alpha_2 f_2 + \cdots + \alpha_k f_k$

$$\min_{\alpha \in \mathbb{R}^k} \| y - \Phi \alpha \|_2^2$$

Suppose $\hat{\alpha} \in \mathbb{R}^k$ is a minimizer, $\hat{\alpha} \in \operatorname{argmin} \| y - \Phi \alpha \|_2^2$

then $\Phi^T \Phi \hat{\alpha} = \Phi^T y$

$\Phi \in \mathbb{R}^{N \times k}$

$$\Phi = \begin{bmatrix} f_1(x_1) & f_2(x_1) & \cdots & f_k(x_1) \\ f_1(x_2) & f_2(x_2) & & f_k(x_2) \\ \vdots & \vdots & & \\ f_1(x_N) & f_2(x_N) & \cdots & f_k(x_N) \end{bmatrix}$$

$$\hat{y}_i = f_1(x_i)\alpha_1 + f_2(x_i)\alpha_2 + \cdots + f_k(x_i)\alpha_k$$

$$= \begin{bmatrix} f_1(x_i) & f_2(x_i) & \cdots & f_k(x_i) \end{bmatrix} \begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \vdots \\ \alpha_k \end{bmatrix}$$

for $i = 1, 2, \dots, N$